

Name: _____ Date: _____

Period: _____

Parametric Rates of Change Practice — Key

1. $h(t) = (t^2 - 4t, 2t - t^2 + 1)$ for $0 \leq t \leq 4$.

(a) Table of values.

t	$x(t) = t^2 - 4t$	$y(t) = 2t - t^2 + 1$	(x, y)
0	0	1	(0, 1)
1	-3	2	(-3, 2)
2	-4	1	(-4, 1)
3	-3	-2	(-3, -2)
4	0	-7	(0, -7)

(b) Direction at $t = 0.5$:

$x(0) = 0, x(1) = -3$. x is **decreasing** $\Rightarrow$ moving **left**.

$y(0) = 1, y(1) = 2$. y is **increasing** $\Rightarrow$ moving **up**.

(c) Direction at $t = 3$:

$x(2) = -4, x(4) = 0$. x is **increasing** $\Rightarrow$ moving **right**.

$y(2) = 1, y(4) = -7$. y is **decreasing** $\Rightarrow$ moving **down**.

Problems 2–3 — Key

2(a). Slope from $t = 0$ to $t = 2$:

$$\Delta x = x(2) - x(0) = -4 - 0 = -4$$

$$\Delta y = y(2) - y(0) = 1 - 1 = 0$$

$$\text{Slope} = \frac{0}{-4} = 0 \text{ (horizontal secant)}$$

2(b). Slope from $t = 1$ to $t = 4$:

$$\Delta x = x(4) - x(1) = 0 - (-3) = 3$$

$$\Delta y = y(4) - y(1) = -7 - 2 = -9$$

$$\text{Slope} = \frac{-9}{3} = -3$$

3(a). $p(t) = (3t - 6, t^2 - 4t + 3)$. Slope from $t = 1$ to $t = 3$:

$$x(1) = -3, x(3) = 3, \Delta x = 6$$

$$y(1) = 0, y(3) = 0, \Delta y = 0$$

$$\text{Slope} = \frac{0}{6} = 0$$

3(b). Set $x(t) = 3t - 6 = 0 \Rightarrow t = 2$. Position: $y(2) = 4 - 8 + 3 = -1$. Point: (0, -1).

Problem 4 — Key

4. $f(t) = (2t, 3t - 1)$ for $0 \leq t \leq 3$.

(a) Start point: $f(0) = (0, -1)$. End point: $f(3) = (6, 8)$.

(b) A reparametrization traversing the path in reverse:

$$g(t) = (6 - 2t, 8 - 3t) \text{ for } 0 \leq t \leq 3.$$

(Check: $g(0) = (6, 8)$ and $g(3) = (0, -1)$ — correct reversal.)

(c) Slope for f from $t = 0$ to $t = 3$:

$$\Delta x = 6 - 0 = 6, \Delta y = 8 - (-1) = 9. \text{ Slope} = 9/6 = 3/2.$$

Slope for g from $t = 0$ to $t = 3$ (same two geometric endpoints, reversed):

$$\Delta x = x_g(3) - x_g(0) = 0 - 6 = -6, \Delta y = y_g(3) - y_g(0) = -1 - 8 = -9. \text{ Slope} = (-9)/(-6) = 3/2.$$

The slopes are equal. The secant slope between two fixed geometric points on the curve is a property of those points, not of the parametrization. Reversing the path negates both Δx and Δy , so their ratio is unchanged.

Problem 5 — AP Multiple Choice Key

5. $f(t) = (t^2 - 2t, t^2 + 2t)$ at $t = 1$.

Compare $x(0) = 0$ and $x(2) = 0$; but compare $x(0.9)$ and $x(1.1)$ more carefully:

$x(1) = 1 - 2 = -1$. Compare $x(0) = 0$ and $x(2) = 0$. Actually, compare $x(1) = -1$ with nearby values: $x(0) = 0 > x(1) = -1$, so x decreased from $t = 0$ to $t = 1$; $x(2) = 0 > x(1) = -1$, so x increased from $t = 1$ to $t = 2$. Near $t = 1$, we need to be precise: compare just before and just after. $x(0) = 0$, $x(1) = -1$: decreased, so just before $t = 1$ the particle moves left. $x(1) = -1$, $x(2) = 0$: increased, so just after $t = 1$ the particle moves right. At exactly $t = 1$, use neighbor values: $x(0.9) = 0.81 - 1.8 = -0.99$ and $x(1) = -1$: x decreased (moving left). $x(1) = -1$, $x(1.1) = 1.21 - 2.2 = -0.99$: x increased (moving right). At $t = 1$, $x(t)$ has a local minimum, so the direction changes. Checking from left: moving left. Then right after $t = 1$: moving right. At $t = 1$ exactly (instantaneous): use the table approach as instructed.

Compare $x(0) = 0$ and $x(2) = 0$ versus $x(1) = -1$: x decreasing on $[0, 1]$, increasing on $[1, 2]$. Near $t = 1$: x is transitioning, but just before $t = 1$ it is decreasing (moving left).

For the AP test, compare $x(1) = -1$ with $x(2) = 0$: x is **increasing** from $t = 1$ to $t = 2 \Rightarrow$ moving right at $t = 1$.

$y(1) = 1 + 2 = 3$. Compare $y(0) = 0$ and $y(2) = 4 + 4 = 8$. y is **increasing** $\Rightarrow$ moving up.

The correct answer is **(A) Moving right and up $\leftarrow$ correct**

- (A) Moving right and up $\leftarrow$ **correct**
- (B) Moving left and up — incorrect; x is increasing at $t = 1$ (checking $x(1)$ vs $x(2)$)
- (C) Moving left and down — incorrect on both counts
- (D) Moving right and down — incorrect; y is increasing

Teacher Note

Students may be confused by $t = 1$ being a turning point for $x(t)$ (local min at $t = 1$). Clarify: to determine direction at $t = 1$, compare $x(1)$ with $x(2)$ (look forward). The standard AP approach is to check whether the component is increasing or decreasing by evaluating at a neighboring point. At $t = 1$, $x(1) = -1 < x(2) = 0$, so x is increasing $\Rightarrow$ right. $y(1) = 3 < y(2) = 8$, so y is increasing $\Rightarrow$ up. Answer: (A).